

A Computational View of Medicine

Why AI will matter most where disease becomes a search problem

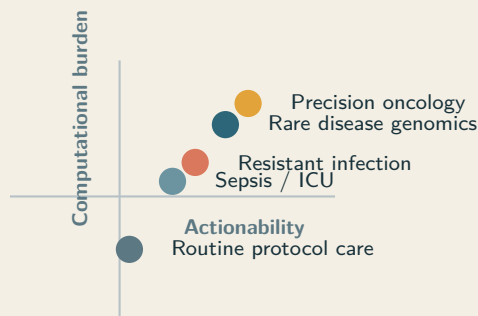
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Thesis. The most important medical use of advanced AI is not generic automation. It is high-context reasoning over difficult, patient-specific problems where better synthesis can change diagnosis, treatment choice, escalation, or trial matching. In these settings, disease is better understood as a computational burden distributed across time, modalities, and decision branches than as a static label alone.

A computational framing Diagnosis as search problem

Where more inference
can improve the next
decision.



Abstract

Artificial intelligence will not improve every part of medicine equally. Its highest-value role is in disease settings where the bottleneck is not data collection or workflow automation but high-context reasoning over multimodal evidence under uncertainty. We argue for a computational view of medicine that treats serious illness as a search problem over longitudinal records, pathology, imaging, genomics, patient-generated data, treatment history, and branching therapeutic options. The key question is not whether a model can produce clinically fluent text, but whether additional inference changes the next real decision for a specific patient. This constraint applies to clinicians and AI systems alike: both face the same search problem, and the quality of the search depends on the depth and structure of reasoning applied. We synthesize evidence from precision oncology, rare-disease genomics, antimicrobial discovery, sepsis surveillance, patient-reported outcomes, open-notes adoption, and N-of-1 medicine. Patient access to records is necessary but insufficient; because clinical reasoning is distributed across specialists and institutions—with the patient as the only persistent entity—data access becomes powerful only when patients and clinicians can repeatedly compute over the full timeline of illness. The result is a framework for directing AI toward cases where reasoning is the bottleneck, outcomes depend on synthesis quality and calibration, and patient-held data can become a force multiplier.

1 Introduction

In many of the cases that matter most in medicine, the limiting factor is not missing data. It is insufficient reasoning over data already available. The difficult work is synthesis: reconciling longitudinal evidence, deciding which signals matter, ranking plausible actions, and updating the model of a case as new information arrives.

This constraint is visible in everyday care. If a clinician had more time to think, would the next decision improve? For many routine cases, probably not. For high-burden cases—a confusing constellation of symptoms, an ambiguous genomic result, a cancer that has stopped responding—the answer is often yes. That makes reasoning a real clinical resource, and it makes reasoning scarcity a real clinical bottleneck.

Clinical reasoning can be carried out by clinicians, AI systems, or human-AI teams. The important question is not who reasons in the abstract, but whether additional reasoning changes the next meaningful decision for a patient. Under this view, advanced AI should be judged not by whether it produces fluent medical language, but by whether it adds useful reasoning capacity where the decision is difficult, high-stakes, and patient-specific. This extends Topol's case for human-machine medicine by placing computational leverage, rather than generic augmentation, at the center [34].

The constraint is not unique to machines. Physicians employ both rapid pattern recognition and slower deliberate analysis—dual-process reasoning—switching between modes depending on case complexity [7]. But time per patient is finite. Approximately 5% of US adults—over 12 million people—experience a diagnostic error in outpatient care each year [31], and an estimated 795,000 Americans suffer permanent disability or death annually from diagnostic errors, concentrated in cardiovascular events, infections, and cancer [24]. The “diagnostic timeout” in emergency medicine, a structured pause to reconsider the differential, reduces diagnostic error [7, 10]. Primary care consultation lengths range from 5 minutes to over 20 across healthcare systems [14]; if reasoning time is part of the bottleneck, that variation should predict quality.

Reasoning-focused AI systems can already participate in this deliberate layer of medicine. They can generate hypotheses, weight differentials, and integrate distributed evidence [22]. On complex diagnostic cases from the *New England Journal of Medicine*, GPT-4 included the correct diagnosis in 64% of cases, compared to 36% for medical-journal readers [19]. Med-PaLM 2 achieved 86.5% on USMLE-style questions and was preferred over physician answers on eight of nine clinical axes [32]. These benchmark results do not prove

clinical utility on their own. They do show that added machine inference can improve the quality of clinical synthesis in some settings.

Reasoning as a clinical resource. The core question in advanced medical AI is not whether machines can reason in the abstract, but whether more reasoning—human, machine, or combined—changes the next real decision for a patient.

We call this a computational view of medicine. It treats disease management as the problem of converting fragmented, multimodal, longitudinal evidence into better patient-specific decisions under uncertainty. The argument of this paper is simple: disease classes differ sharply in how much more computation can improve outcomes, patient-held data changes where that computation can happen, and the highest-value uses of AI will concentrate in the cases where reasoning is the bottleneck.

Definition. A *computational view of medicine* treats disease management as the problem of converting fragmented, multimodal, longitudinal evidence into better patient-specific decisions under uncertainty.

2 A framework for computational leverage

We propose five properties that determine whether a disease class is likely to benefit from larger inference budgets:

1. **Search-space size:** how many plausible hypotheses, targets, interventions, or sequences must be evaluated.
2. **Data burden:** how many modalities and how much longitudinal context are required to reason well.
3. **Personalization:** how much the correct answer depends on the specific patient rather than the average protocol.
4. **Actionability:** whether better synthesis can materially alter diagnosis, escalation, treatment choice, or trial eligibility.
5. **Compute elasticity:** whether additional reasoning depth is likely to improve the result instead of merely restating the same answer more fluently.

These variables describe a single question: when does more reasoning change the next decision? Precision oncology, rare-disease genomics, resistant infection, critical care syndromes, and long-horizon chronic conditions rise to the top because they combine large search spaces, heterogeneous data, and decisions whose quality directly affects outcome. Diseases governed primarily by narrow protocols benefit more from reliable automation than from deeper inference.

Compute elasticity is the key discriminator. In low-elasticity domains, a clinician at normal pace already reaches the correct decision and more thought adds little. In high-elasticity domains, additional reasoning can shift the differential, reorder treatment options, improve calibration, or surface a branch that would otherwise be missed. AI does not replace clinical reasoning in these cases. It extends the reasoning budget available before action is taken.

Domain	Search	Data	Pers.	Action	Elast.	Interpretation
Precision oncology	5	5	5	5	5	Dense multimodal evidence and many branching intervention paths make more inference directly decision-relevant.
Rare-disease genomics	5	5	5	4	5	Large hypothesis spaces and sparse distributed clues create unusually high value for long-context reasoning.
Resistant infection	4	4	4	5	4	The answer depends on organism, host state, prior exposure, and time-sensitive escalation.
Sepsis / ICU	4	5	4	5	4	Continuous streaming data create a high-burden monitoring problem in which timing dominates outcome.
Chronic inflammatory disease	3	4	4	4	3	Longitudinal symptom and biomarker integration can improve timing and sequencing, but action spaces are narrower.
Routine protocol care	1	2	1	2	1	Most value lies in reliable execution and logistics rather than larger inference budgets.

Table 1: Illustrative scoring matrix for computational leverage. Scores are conceptual rather than empirical, but they make the framework explicit and falsifiable.

This is where the abstract framework meets clinical work. A molecular tumor board may have minutes to search a decision space of drugs, trials, biomarkers, and prior-line contingencies. A geneticist may have one review session to integrate years of notes, family history, imaging, and genome data. A primary care physician managing multimorbidity must often act before the full interaction space has been searched. We refer to this gap between what could in principle be searched and what is actually searched in routine care as the *compute deficit*. The largest returns from medical AI will come from shrinking that gap in domains where better search changes patient care.

Table 1 should therefore be read as a proposed allocation heuristic, not as a finished empirical scorecard. It makes the framework explicit enough to test. *Search-space size* can be approximated by counting plausible differentials or candidate interventions; *data burden* by the number of modalities and timescales required for a defensible decision; and *compute elasticity* by asking whether more reasoning time improves ranking quality, calibration, or branch selection. Figure 1 addresses a different part of the argument: not leverage itself, but urgency. The concentration of serious diagnostic harm in the same broad domains suggests that the reasoning gap is a measurable public-health problem rather than a theoretical one.

Why prevalence is not enough. Common diseases are not automatically the best targets for large inference budgets. The relevant question is whether more synthesis can change a meaningful patient-specific decision. That is why prevalence rankings and compute-opportunity rankings diverge.

3 Why patient-held data matter

Clinical reasoning in modern healthcare is distributed across general practitioners, specialists, pathologists, radiologists, and multidisciplinary teams. Each referral adds expertise, but each handoff also risks losing state. The patient is the only entity that persists across all of those encounters. That fact matters computationally. If the case is going to remain coherent across time and institutions, the longitudinal state of the problem has to be reconstructable around the patient.

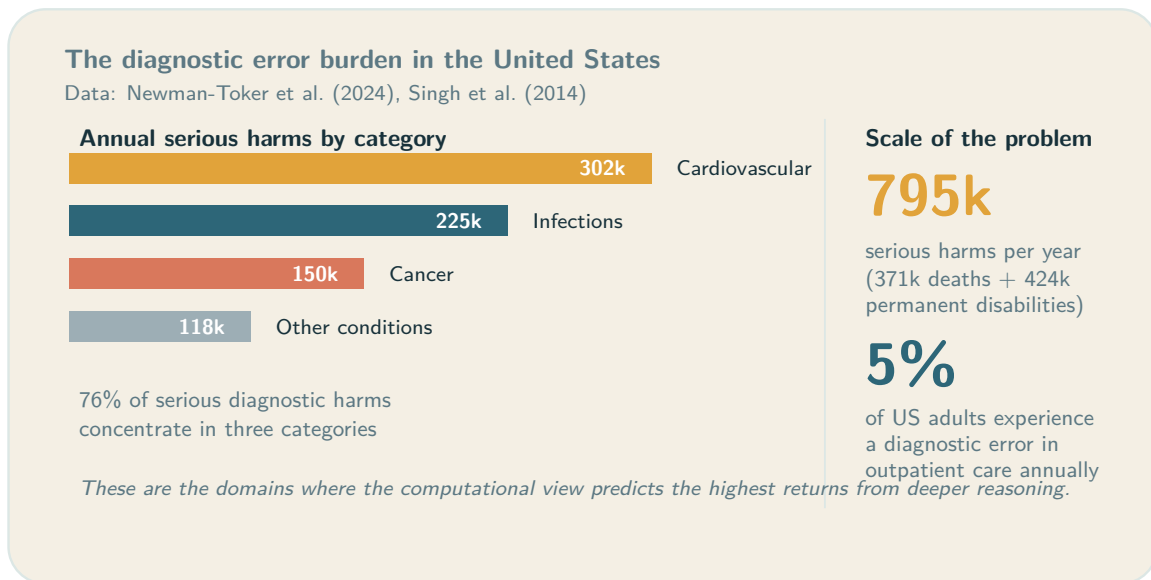


Figure 1: The diagnostic error burden concentrates in exactly the domains where the computational view predicts the highest compute elasticity. Data from Newman-Toker et al. [24] and Singh et al. [31].

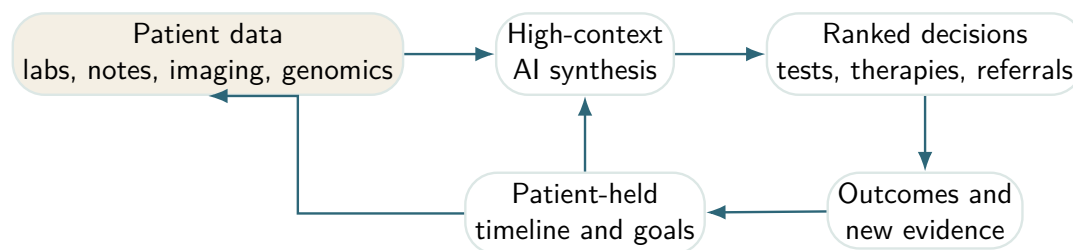


Figure 2: The patient-compute loop. In high-burden disease, value comes from repeated synthesis over the full timeline rather than one-shot question answering.

This is why patient-held data matters. It is not only a matter of access rights or transparency. It changes where medical reasoning can live. Once records can be assembled across encounters, institutions, and modalities, computation can follow the patient rather than remaining trapped inside any one clinic, hospital, or specialty workflow.

OpenNotes showed that patients value direct access to clinicians' notes and that transparency can improve engagement [8]. Patients also shared those notes with family members and caregivers, turning the record into a coordination tool rather than a private institutional artifact [15]. Patient-generated data—symptom tracking, wearables, and home monitoring—extend that record beyond formal encounters [25]. The longitudinal case history is therefore no longer only what the institution stores. It is increasingly what can be assembled around the patient over time.

Access alone, however, is not enough. Precision medicine remains incomplete without patients as participants in how data are interpreted and used [35]. AI for shared decision-making must improve deliberation, not merely generate recommendations [12]. Integration of patient-generated health data is not only a technical problem but a clinical and organizational one [16]. Early work suggests that patients can already use large language models on open notes to interpret their records [29]. The larger implication is architectural: without patient-held continuity of state, longitudinal computation cannot reliably follow the case; with it, repeated reasoning over the full illness trajectory becomes possible.

4 Evidence from the highest-leverage domains

4.1 Precision oncology

Cancer is already a computational problem. Molecular diagnosis, resistance interpretation, treatment sequencing, and trial matching require synthesis across pathology, imaging, genomics, prior therapy, and time. Personalized cancer vaccines make this explicit: identifying patient-specific tumor mutations, ranking neoantigens, and manufacturing a bespoke intervention. Sahin and colleagues showed that individualized RNA mutanome vaccines mobilize poly-specific immunity against melanoma [28]. AlphaFold expands the search surface further by making protein consequence analysis tractable across large molecular spaces [18].

Compute sits inside the decision itself: which mutation matters, which escape mechanisms are plausible, which combination is worth pursuing, which trial is defensible for this patient now. The burden of reasoning has become part of the disease burden.

A single patient may generate surgical pathology, radiology progression patterns, panel sequencing, circulating tumor DNA, treatment toxicities, and eligibility-relevant comorbidities. The computational task is to rank drivers, resistance hypotheses, and therapeutic branches while making uncertainty visible. The leverage comes not from generating prose about cancer but from compressing a large search problem into a smaller set of defensible next moves.

4.2 Rare disease genomics

The problem is not only finding a variant but aligning phenotype, pedigree, prior negative workup, and the literature fast enough to affect care. Wojcik and colleagues showed in the *New England Journal of Medicine* that genome sequencing materially improved diagnosis for suspected rare disease [36]. This is a canonical compute-heavy domain: large hypothesis spaces, sparse signal, fragmented records, and a direct payoff when reasoning succeeds. The cost of failure is now quantified: rare-disease patients accumulate average healthcare costs of €27,000 over their diagnostic odyssey, compared to €3,600 for matched controls, and avoidable costs of delayed diagnosis in the US range from \$86,000 to over \$517,000 per patient [9].

4.3 Resistant infection and antibiotic discovery

Antimicrobial resistance presents a two-layer search problem. The first is patient-specific: organism identity, susceptibility, host state, prior exposure, organ function, and drug interactions. The second is discovery: searching chemical space for compounds with novel activity. Stokes and colleagues used deep learning to identify halicin, accelerating antibiotic discovery in regions of chemical space invisible to standard pipelines [33]. More compute means better regimen ranking in the clinic and better exploration of candidate space in the lab.

4.4 Sepsis and critical care

In critical care the challenge is time. Dozens of measurements update continuously, and the relevant signal is distributed across trends rather than single observations. A machine-learning sepsis early-warning system deployed across multiple hospitals was associated with faster antibiotic administration and lower mortality [1]. Deep learning on electronic health records can scale across broad clinical prediction tasks using the raw longitudinal record [27], and pragmatic deterioration surveillance can attach to live escalation pathways

[23]. These results establish that longitudinal records contain actionable signal, and systems that hold more context together improve timing-sensitive decisions.

4.5 Psychiatry and mental health

Psychiatry is computational because the organ of interest is computationally complex. The relevant clinical state is not captured by a single laboratory value or imaging slice; it is distributed across history, context, behavior, sleep, substance use, prior treatment response, and the patient's own account of what changes their internal state. A clinician deciding what to do next in depression, bipolar disorder, ADHD, or anxiety is therefore working with a partially observed system whose state has to be inferred from a noisy longitudinal record.

That makes psychiatric care, in a concrete sense, a sequential decision problem. Each intervention is an action taken under uncertainty: start a medication, stop one, increase a dose, change a therapy target, alter environmental structure, test a behavioral strategy, or wait for more signal. The next state of the patient depends partly on that action and partly on dynamics that are difficult to observe directly. When care works well, whether in psychotherapy, pharmacotherapy, or a combination, it is because the clinician and patient find a better next action for the current state.

This is visible in pharmacotherapy. First-line antidepressants fail in a large share of patients, and standard care often proceeds by serial trial and error over months. Pharmacogenomics offers one example of computation reducing that search burden: cytochrome P450 variation can identify medications that are more likely to be poorly metabolized before they are prescribed [13]. The point is not that genotype solves psychiatry. It is that psychiatric treatment already contains a branching search problem in which better synthesis can eliminate bad branches earlier.

The same structure appears in psychotherapy. Cognitive behavioral therapy is, in practice, an effort to map the loops that keep a disorder stable: triggers, beliefs, behaviors, and reinforcing consequences [5]. That is close to an action-selection problem over state. A therapist is trying to identify which intervention is most likely to move the patient from a maladaptive local equilibrium toward a better one, given the current pattern of thought, behavior, and context. Under a computational view, psychiatry belongs in the portfolio of high-burden reasoning problems precisely because more context, better state estimation, and better action selection can change the next therapeutic move.

4.6 Patient-reported outcomes and symptom monitoring

Not all high-value computation depends on genomics. Electronic patient-reported symptom monitoring during cancer treatment improved overall survival, likely by enabling earlier response to deterioration patterns otherwise missed [3, 4]. Symptoms, side effects, and tolerability live outside the hospital until severe enough to trigger contact. In the computational paradigm, those signals become first-class inputs into the case model.

5 N-of-1 medicine and personal computation

The computational view converges with N-of-1 medicine. Schork argued that precision medicine must be understood at the level of the individual, not the cohort average [30]. Guyatt laid the foundation for randomized trials in individual patients [11]; Lillie argued that N-of-1 trials align with the actual endpoint of clinical practice: optimizing care for one person [20]. Augustine proposed a roadmap for N-of-1 therapies

Domain	Core data	How computation is leveraged	Representative evidence
Precision oncology	Genomics, pathology, imaging, treatment history	Variant interpretation, resistance modeling, neoantigen ranking, regimen sequencing, trial matching	Sahin et al. [28]; Jumper et al. [18]
Rare disease	Phenotype timeline, pedigree, genome sequencing, prior workup	Cross-linking phenotype ontologies, literature, family history, and candidate variants	Wojcik et al. [36]
Resistant infection	Microbiology, susceptibilities, organ function, prior antibiotic exposure	Better ranking of therapies and faster search across compound or regimen space	Stokes et al. [33]
Sepsis / ICU	Streaming vitals, labs, notes, medication events	Earlier escalation from weak, time-distributed patterns across the longitudinal record	Adams et al. [1]; Rajkomar et al. [27]; Moses et al. [23]
Patient-reported symptoms	Patient-reported outcomes, treatments, toxicities, goals	Detecting deterioration and treatment intolerance before they become acute	Basch et al. [3]; Basch et al. [4]
Patient-held records	Notes, meds, labs, imaging reports, symptoms, goals	Converts raw data access into usable reasoning leverage before and between encounters	Delbanco et al. [8]; Salmi et al. [29]

Table 2: High-value disease areas under a computational lens.

covering scientific, operational, and regulatory requirements [2]; Jonker outlined how N-of-1 medicine could reshape treatment evaluation for rare and heterogeneous disease [17]. These contributions move personalization from slogan to organizing scientific unit.

Longitudinal self-measurement provides parallel evidence. Integrative personal omics profiling reveals dynamic molecular phenotypes within a single individual over time [6]. Dense personal data clouds can monitor health transitions with temporal resolution far exceeding episodic care [26]. Biology, symptoms, and risk are often more legible in longitudinal within-person data than in sparse snapshots.

Public self-experimentation—highly instrumented programmes such as Bryan Johnson’s—is not clinical evidence. What it reveals is latent demand: people want to run dense longitudinal data loops over their own bodies. The scientific path forward is audited N-of-1 workflows combining patient goals, validated measurements, and clinically accountable reasoning.

6 Case examples

Table 2 shows why a single computational model fits these otherwise different domains.

Future patient personas		
Patient	Data substrate	Computational leverage
Asha, 58 metastatic NSCLC	Tumour DNA/RNA, pathology, CT trends, prior lines, toxicities, performance status	Variant prioritization, resistance hypotheses, neoantigen ranking, and live trial matching change the next therapeutic branch.
Mateo, 6 undiagnosed neuro- developmental syndrome	Phenotype timeline, EEG, MRI, family history, genome sequencing, prior negative workup	Phenotype-variant ranking and literature synthesis narrow candidate mechanisms and guide targeted confirmatory testing.
Leila, 34 Crohn's disease on biologics	Symptoms, calprotectin, medication adherence, wearables, sleep, diet, clinician notes	Within-person pattern detection turns fragmented self-tracking into actionable flare prediction and better-timed escalation.

Figure 3: Concrete personas for a computational health stack. The same design principle appears across specialties: assemble the full timeline, compute over it repeatedly, and use the result to improve the next decision.

6.1 Concrete pathways

In each case, computation improves preparation, ranking, and timing for the next branch in care—not to replace the clinician but to strengthen the decision.

1. **Precision oncology.** A patient-assembled case file combines sequencing, pathology, imaging, and prior therapies. Compute ranks drivers, resistance mechanisms, toxic trade-offs, and live trials. The output is a defensible next branch with explicit uncertainty.
2. **Rare disease.** A phenotype timeline with milestones, regressions, imaging, and prior exclusions is linked to genome evidence and literature, eliminating time spent re-deriving the case at each referral.
3. **Chronic inflammatory disease.** Symptoms, labs, wearable data, and medication exposure are integrated within-person to identify precursors of flare, intolerance, or non-response.

7 Evaluation and endpoints

The computational view should change how health AI is evaluated. Discrimination metrics are not enough. In high-stakes medicine, calibration matters as much as accuracy: a system that outputs “60% diagnosis A, 25% diagnosis B, and here is the test that discriminates” is more useful than a bare point estimate, even at identical accuracy. More reasoning sometimes does not change the top-ranked answer but sharpens the uncertainty estimate, which clinically determines whether a clinician orders a confirmatory test, refers, or treats empirically. The relevant question is whether synthesis quality—including calibration—improved at the point where the case could still change direction.

Study type	What changed	Why it matters for the thesis
Pragmatic randomized clinical trial in cardiovascular screening	AI-enabled ECG alerts were tested as an intervention rather than a retrospective classifier [21]	Supports the claim that computational systems can change downstream decisions and outcomes, not merely generate scores.
Pragmatic cluster-randomized deterioration trial	Real-time surveillance for patient deterioration was tested across live inpatient settings [23]	Strengthens the case that high-context clinical surveillance can function as an operational care system.
Patient-record sharing study	Patients with transparent notes shared them with family and caregivers, extending the practical reach of the record [15]	Shows that once patients have access to records, those records begin acting as coordination infrastructure.
Patient-side LLM proof of concept	Patients used large language models on open notes to better understand and work with their records [29]	Gives direct support to the argument that personal compute can become part of the patient toolkit.

Table 3: Evidence that the computational view extends beyond retrospective prediction.

8 What should be built

The practical implication is narrower than much current health-AI rhetoric. The field does not primarily need more conversational surfaces. It needs reasoning infrastructure: systems that assemble patient records across sites, preserve chronology, separate evidence from inference, and return ranked next steps rather than fluent summaries. The right unit is a living case model that can be re-run as disease evolves.

These systems should strengthen the clinical encounter rather than bypass it. The design target is a patient-clinician conversation with better state, better uncertainty handling, and better preparation before the next branch in care is chosen.

9 Risks and constraints

The computational view sharpens the need for evidence and oversight. High-burden disease is exactly where hallucinated synthesis, hidden bias, and brittle prompting are most dangerous. Reasoning systems must therefore be auditable, source-grounded, and explicit about what they do not know.

Not every disease is compute-elastic. Some problems remain dominated by biology, logistics, or protocol execution rather than reasoning depth. And if compute meaningfully improves outcomes, unequal access to compute becomes a health inequity. A health stack that gives patients records without usable synthesis leaves too much leverage with the institution.

10 Operational consequences

The agenda for health AI changes. Clinical systems should separate protocol problems from search problems. Regulators should ask not only whether a model is accurate on average, but whether it improves reasoning

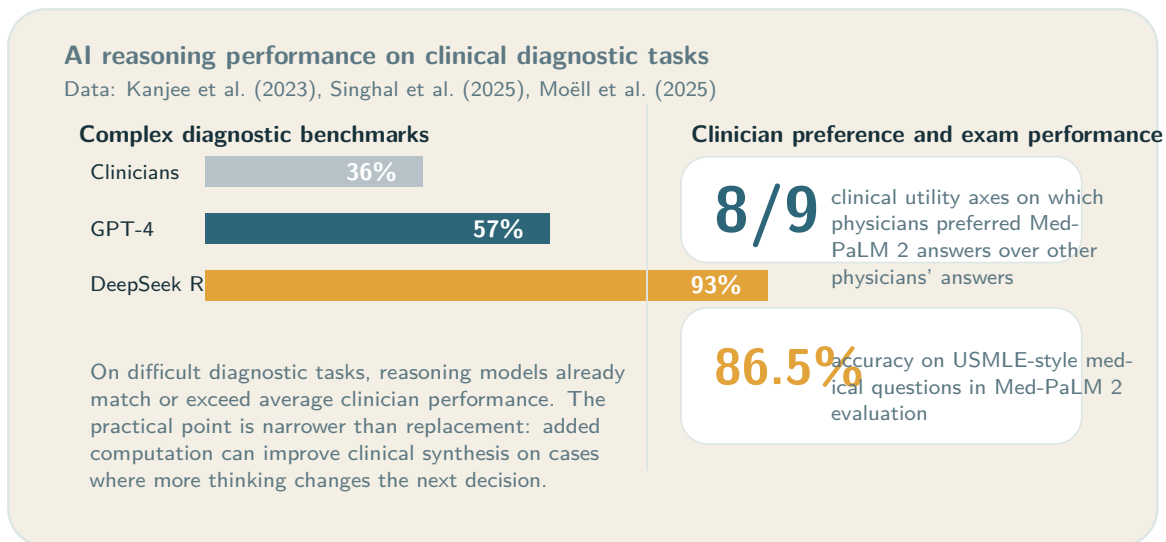


Figure 4: Reasoning-focused AI systems already perform strongly on difficult diagnostic and medical benchmarks. Data from Kanjee et al. [19], Singhal et al. [32], and Moëll et al. [22]. The clinical relevance is practical: in some cases, more computation improves the quality of synthesis.

quality in the cases where reasoning is the bottleneck. And product teams should direct the largest inference budgets toward cases where better synthesis can change the next branch in care.

The practical payoff is concentration: fewer missed driver mutations, faster rare-disease diagnoses, better antibiotic choices, earlier sepsis escalation, and better continuity across fragmented care. Pragmatic intervention studies in ECG alerting, deterioration surveillance, and symptom monitoring suggest that this shift can move from theory to operations when attached to real decision paths [3, 21, 23].

11 Conclusion

Medicine is already a computational process. Evidence-based medicine formalized that process at the population level. The computational view proposed here formalizes it at the patient level and argues that a large share of avoidable harm in complex disease comes from the gap between what could be searched, synthesized, and updated for a patient and what is actually computed in routine care.

The diseases that matter most under this view are not simply the most prevalent. They are the ones that become search problems at patient resolution: precision oncology, rare-disease genomics, resistant infection, selected critical care settings, and chronic relapsing disease. In these domains, better reasoning can change the next branch in care.

The deeper implication is structural. If reasoning has historically been scarce and expensive—bound to the time, memory, and attention of individual clinicians—then cheap, portable computation changes not only the quality of individual decisions but the architecture of medicine itself. A computational view of medicine therefore does two things at once: it shows where additional inference can genuinely improve outcomes, and it points toward a health system in which understanding becomes a shared, longitudinal, patient-held capability rather than a privilege of the institution.

Domain	Primary endpoint candidates	Why these endpoints fit the computational thesis
Rare disease	Time to diagnosis; diagnostic yield; avoided procedures	Measures whether synthesis reduced time spent on the wrong branch of care.
Precision oncology	Time to molecular recommendation; trial-match yield; change in treatment ranking; time to next line	Measures whether computation improved branching decisions rather than documentation alone.
Sepsis / ICU	Time to escalation; antibiotics or intervention timing; false alerts per patient-day; mortality or ICU transfer	Measures whether longitudinal synthesis improved timing without overwhelming workflow.
Chronic disease / N-of-1 care	Time to flare detection; within-person treatment response; symptom burden; adherence to preferred regimen	Measures whether repeated computation improves individualized management over time.
Patient-facing record synthesis	Patient comprehension; visit preparedness; question quality; fewer duplicated tests; continuity across sites	Measures whether access plus reasoning improved the practical utility of records.

Table 4: Evaluation endpoints aligned with a computational view of medicine. The goal is to measure whether reasoning improves decisions, not only whether predictions look accurate offline.

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